Chapter 1 Hilbert Space

Chapter 2 Semi-Norms

2.1 Semi-Norms and Locally Convex Linear Topological Spcaes

2.2 Space $D(\Omega)$

Definition 2.2.0.1 Take the following notation:

- $C_0^\infty(\Omega)$ with a $\mathbb{R}$-module structure;
- $D_k(\Omega) := \{f \in C_0^\infty | supp(f) \subseteq K\}$.

Define the seminorm on $D_K(\Omega)$ by:

$$p_{k,m}(f) := \sup_{|s|\leq m, x\in K} |D^s f(x)|, m < \infty$$

Remark The contents, which is related to 'fundamental definition and basic properties of xxx space', will be given(in principle, except metric space?) 'Topological Vector Space'.

Chapter 3 title